

GATE Practice: C Data Types & Operators

35 questions • Questions first • Answer key on the last page

1. Data Types

Which of the following statements about C data types is correct?

- A) char is always 16 bits
- B) int is always exactly 32 bits
- C) short is at least 16 bits and long is at least 32 bits
- D) long can be smaller than short

2. Operators

What is the value of x after executing?

```
int x = 10;  
x += 3 * 2;
```

- A) 16 B) 26 C) 36 D) 60

3. Unary Operator

What is the output?

```
int x = 5;  
printf("%d", -x);
```

- A) 5 B) -5 C) 0 D) Error

4. Pre-increment

What is the output?

```
int x = 5;  
int y = ++x;  
printf("%d %d", x, y);
```

- A) 5 6 B) 6 5 C) 6 6 D) 5 5

5. Post-increment

What is the output?

```
int x = 5;  
int y = x++;  
printf("%d %d", x, y);
```

- A) 5 6 B) 6 5 C) 6 6 D) 5 5

6. Mixed Increment

What is the value of x?

```
int x = 5;  
x++;  
++x;  
x--;
```

A) 4 B) 5 C) 6 D) 7

7. Relational Operators

What is the value of x?

```
int x = 10 > 5;
```

A) 10 B) 5 C) 1 D) 0

8. Logical AND

What is the output?

```
int x = 10;  
printf("%d", (x > 5) && (x < 20));
```

A) 0 B) 1 C) 10 D) 20

9. Logical OR

What is the output?

```
int x = 10;  
printf("%d", (x < 5) || (x == 10));
```

A) 0 B) 1 C) 5 D) 10

10. Logical NOT

What is the output?

```
int x = 0;  
printf("%d", !x);
```

A) 0 B) 1 C) -1 D) Error

11. Logical NOT — GATE Style

What is the output?

```
int x = 10;  
printf("%d", !!x);
```

A) 0 B) 1 C) 10 D) -10

12. Short-Circuit AND

What is the output?

```
int x = 0;  
if (x != 0 && 10 / x > 2) printf("A"); else printf("B");
```

A) A B) B C) Runtime division-by-zero error D) Compilation error

13. Short-Circuit OR

What is the output?

```
int x = 10;  
if (x == 10 || ++x > 20) printf("%d", x);
```

A) 10 B) 11 C) 20 D) 21

14. Short-Circuiting

What is the final value of x?

```
int x = 5;  
(1 || ++x);
```

A) 5 B) 6 C) 1 D) Undefined

15. Ternary Operator

What is the output?

```
int a = 10, b = 20;  
int x = (a > b) ? a : b;  
printf("%d", x);
```

A) 10 B) 20 C) 30 D) 0

16. Nested Ternary

What is the output?

```
int a = 5, b = 10, c = 15;  
int x = (a > b) ? a : (b > c) ? b : c;  
printf("%d", x);
```

A) 5 B) 10 C) 15 D) 0

17. Bitwise AND

What is the value of 5 & 3?

A) 1 B) 2 C) 7 D) 8

18. Bitwise OR

What is the value of 5 | 3?

A) 1 B) 6 C) 7 D) 8

19. Bitwise XOR

What is the value of $5 \wedge 3$?

- A) 1 B) 2 C) 6 D) 7

20. Bitwise NOT

Assuming a 32-bit signed two's-complement integer, what is the value of ~ 5 ?

- A) 4 B) -5 C) -6 D) 6

21. Left Shift

What is the value of $3 \ll 2$?

- A) 5 B) 6 C) 8 D) 12

22. Right Shift

Assuming x is a positive integer, what is the value of $20 \gg 2$?

- A) 2 B) 4 C) 5 D) 10

23. Bitwise vs Logical

What is the output?

```
printf("%d %d", 5 && 3, 5 & 3);
```

- A) 1 1 B) 1 7 C) 1 2 D) 5 3

24. Operator Precedence

What is the output?

```
int x = 2 + 3 * 4;  
printf("%d", x);
```

- A) 20 B) 14 C) 24 D) 9

25. GATE-Style Expression

What is the output?

```
int a = 5, b = 2;  
printf("%d", a++ + ++b);
```

- A) 7 B) 8 C) 9 D) 10

26. GATE-Style

What is the final value of x ?

```
int x = 5;  
x = x++ + 2;
```

- A) 5 B) 6 C) 7 D) The behavior is undefined

27. Bitwise Combination

What is the value of $8 | 4 \& 2$?

A) 0 B) 2 C) 8 D) 14

28. Shift + Bitwise

What is the value of $(5 \ll 1) | 2$?

A) 7 B) 10 C) 12 D) 14

29. Logical + Relational

What is the output?

```
int a = 5, b = 10, c = 15;
printf("%d", a < b && b < c);
```

A) 0 B) 1 C) 5 D) 15

30. GATE-Level Mixed Question

What is the output?

```
int a = 4, b = 3;
int x = a++ & ++b;
printf("%d %d %d", a, b, x);
```

A) 5 4 0 B) 5 4 4 C) 4 4 4 D) 5 3 7

31. GATE-Level Short Circuit

What is the final value of x?

```
int x = 5;
if ((x > 10) && (++x > 5)) x++;
printf("%d", x);
```

A) 5 B) 6 C) 7 D) 10

32. GATE-Level Logical + Increment

What is the output?

```
int x = 0;
if (x++ || ++x) printf("%d", x); else printf("%d", x);
```

A) 0 B) 1 C) 2 D) Undefined behavior

33. Bit Manipulation

What is the value of x?

```
int x = 12;  
x = x >> 1;  
x = x << 2;
```

A) 6 B) 12 C) 24 D) 48

34. XOR Property

What is the value of $7 \wedge 7$?

A) 0 B) 1 C) 7 D) 14

35. GATE-Style Multiple Select

Which statements are true in C?

- A) && performs logical AND
- B) & performs bitwise AND
- C) || performs logical OR
- D) | performs bitwise OR

Select all correct options.

ANSWER KEY

Check your answers only after attempting all 35 questions.

1-C 2-A 3-B 4-C 5-B

6-C 7-C 8-B 9-B 10-B

11-B 12-B 13-A 14-A 15-B

16-C 17-A 18-C 19-C 20-C

21-D 22-C 23-C 24-B 25-B

26-D 27-C 28-A 29-B 30-B

31-A 32-C 33-B 34-A 35-A, B, C, D